

EASTERN IOWA, IA, USA

WMO: 725450

Lat: 41.883N

Lon: 91.717W

Elev: 265

StdP: 98.19

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14990

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.6	-19.6	-26.0	0.4	-22.3	-23.2	0.5	-19.2	14.5	-6.3	12.7	-8.7	4.4	310	0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	32.3	24.6	30.9	23.7	29.3	22.8	26.1	30.4	25.1	29.1	24.1	27.9	4.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.9	20.6	29.0	23.8	19.3	27.8	22.8	18.1	26.6	82.5	30.3	77.8	29.0	73.7	28.0	31.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.0	10.7	9.2	-27.1	34.5	3.3	1.8	-29.5	35.8	-31.4	36.9	-33.3	37.9	-35.7	39.2		
(5)			-27.3	27.8	3.3	1.0	-29.7	28.6	-31.6	29.2	-33.4	29.7	-35.8	30.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.3	-6.7	-4.4	2.5	9.6	15.8	21.1	22.8	21.7	17.8	10.8	3.2	-3.6		
	DBStd	11.67	7.18	7.07	6.58	5.35	4.64	3.45	3.09	2.96	4.45	5.38	5.79	6.64		
	HDD10.0	1938	518	404	246	69	7	0	0	0	2	55	214	423		
	HDD18.3	3732	776	637	491	265	107	13	3	6	62	239	453	680		
	CDD10.0	1682	0	1	14	58	186	333	397	363	237	81	11	1		
	CDD18.3	435	0	0	1	3	28	97	142	110	48	7	0	0		
	CDH23.3	3781	0	0	5	54	249	830	1280	874	424	64	1	0		
	CDH26.7	1152	0	0	0	11	58	259	434	251	126	14	0	0		
(14) Wind	WSAvg	4.4	5.1	5.1	5.2	5.4	4.7	3.8	3.2	3.0	3.5	4.3	4.8	4.7		
(15) Precipitation	PrecAvg	873	26	25	55	84	100	116	115	103	94	58	52	40		
	PrecMax	1536	66	84	155	145	220	293	433	332	260	141	164	91		
	PrecMin	490	1	2	4	18	9	4	11	11	10	1	4	6		
	PrecStd	232	17	17	35	34	53	68	86	74	59	35	36	23		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.9	15.3	23.8	28.4	31.1	32.9	35.1	33.8	32.3	28.5	21.2	15.0		
		MCWB	7.2	10.5	16.2	18.0	21.9	24.5	26.0	24.8	23.1	20.6	14.4	13.1		
	2%	DB	7.0	11.1	19.3	24.6	28.1	31.4	32.4	31.2	29.9	24.9	17.3	10.9		
		MCWB	4.8	7.7	13.2	15.9	19.7	23.4	25.7	24.6	22.2	17.7	12.7	8.9		
	5%	DB	4.1	7.4	16.0	21.7	26.1	29.7	30.7	29.3	27.9	22.2	14.4	7.6		
		MCWB	2.4	5.1	11.9	14.2	18.5	22.5	24.6	23.6	20.9	16.8	10.8	5.8		
	10%	DB	2.1	4.6	12.3	18.8	24.1	27.9	29.0	27.7	25.8	19.5	11.9	4.7		
		MCWB	1.0	2.9	8.3	12.5	17.6	21.5	23.4	22.4	19.6	14.8	9.0	3.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.9	11.9	17.0	19.1	23.1	26.0	28.0	27.0	25.0	21.8	15.8	13.7		
		MCDB	9.4	14.7	22.4	25.4	28.8	31.1	32.2	31.5	29.5	26.2	19.0	15.2		
	2%	WB	4.7	8.0	14.9	17.0	21.6	24.9	26.7	25.6	23.5	19.6	13.6	9.0		
		MCDB	7.0	10.6	18.1	22.4	26.4	29.7	30.8	29.6	27.8	23.0	16.4	10.2		
	5%	WB	2.6	5.2	11.9	15.4	20.2	23.7	25.5	24.6	22.2	17.7	11.6	5.6		
		MCDB	4.0	7.3	15.7	20.6	24.4	28.1	29.3	28.1	26.1	20.3	14.1	7.3		
	10%	WB	1.0	2.9	8.7	13.3	18.7	22.6	24.4	23.5	20.9	15.6	8.9	3.2		
		MCDB	1.9	4.6	11.9	17.9	22.6	26.6	28.1	26.6	24.4	19.2	11.4	4.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.7	9.1	10.3	11.7	11.1	10.6	10.6	10.7	12.1	11.2	9.9	8.5		
		MCDBR	11.5	12.1	14.7	16.2	13.3	12.2	11.5	11.9	13.7	13.9	13.3	11.4		
	5% WB	MCWBR	9.1	8.9	9.2	9.1	6.9	5.9	5.9	6.2	6.6	7.8	9.1	9.0		
		MCWBR	10.5	11.8	13.2	13.9	11.5	10.8	10.5	10.5	11.4	11.3	12.1	10.5		
(40) Clear-Sky Solar Irradiance	taub		0.286	0.304	0.331	0.377	0.416	0.422	0.436	0.416	0.385	0.338	0.299	0.286		
	taud		2.400	2.361	2.390	2.278	2.235	2.268	2.213	2.277	2.326	2.460	2.494	2.427		
	Ebn at Noon		866	903	916	891	861	854	837	843	845	854	848	834		
	Edh at Noon		79	96	106	128	137	133	139	127	112	87	72	71		
(44) All-Sky Solar Radiation	RadAvg		1.83	2.83	3.79	4.69	5.54	6.14	6.17	5.41	4.55	2.99	2.01	1.48		
	RadStd		0.20	0.25	0.35	0.38	0.39	0.57	0.42	0.34	0.35	0.35	0.16	0.15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page